

FA2

Awit, Julia Theresa

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Problem 3.1

Given the data matrix

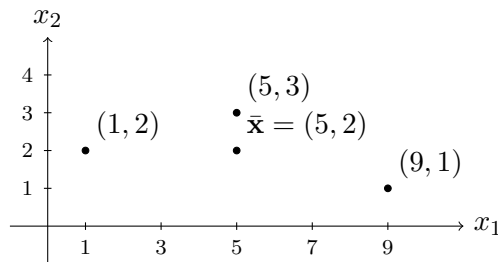
$$X = \begin{pmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{pmatrix}.$$

First, find the sample mean of each column.

$$\bar{x}_1 = \frac{9 + 5 + 1}{3} = 5, \quad \bar{x}_2 = \frac{1 + 3 + 2}{3} = 2$$

$$\bar{\mathbf{x}} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

(a) The three points are $(9, 1)$, $(5, 3)$, $(1, 2)$. The sample mean $(5, 2)$ is plotted as the center of the three points.



(b) Now treat each column as a vector in $n = 3$ dimensions.

$$\mathbf{y}_1 = \begin{pmatrix} 9 \\ 5 \\ 1 \end{pmatrix}, \quad \mathbf{y}_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

Subtract the mean (times $\mathbf{1}$) from each vector to get the deviation vectors.

$$\mathbf{y}_1 - \bar{x}_1 \mathbf{1} = \begin{pmatrix} 9 \\ 5 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ -4 \end{pmatrix}$$

$$\mathbf{y}_2 - \bar{x}_2 \mathbf{1} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

These are the deviation vectors $\mathbf{d}_1 = (4, 0, -4)'$ and $\mathbf{d}_2 = (-1, 1, 0)'$. In the sketch, $\mathbf{y}_1$ and $\mathbf{y}_2$ are vectors in $\mathbb{R}^3$, and $\mathbf{d}_1$, $\mathbf{d}_2$ are what is left after removing their projection onto the line through $\mathbf{1} = (1, 1, 1)'$.

(c) Now plot $\mathbf{d}_1$ and $\mathbf{d}_2$ starting from the origin.



Length of $\mathbf{d}_1$:

$$\|\mathbf{d}_1\| = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{32} = \boxed{4\sqrt{2}}$$

Length of $\mathbf{d}_2$:

$$\|\mathbf{d}_2\| = \sqrt{(-1)^2 + 1^2 + 0^2} = \boxed{\sqrt{2}}$$

Dot product:

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = (4)(-1) + (0)(1) + (-4)(0) = -4$$

Cosine of the angle between them:

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|} = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = \frac{-4}{8} = \boxed{-\frac{1}{2}}$$

Relating this to S_n : using $S_n = \frac{1}{n} D' D$,

$$s_{11} = \frac{\|\mathbf{d}_1\|^2}{3} = \frac{32}{3}, \quad s_{22} = \frac{\|\mathbf{d}_2\|^2}{3} = \frac{2}{3}, \quad s_{12} = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{3} = -\frac{4}{3}$$

$$S_n = \boxed{\begin{pmatrix} 32/3 & -4/3 \\ -4/3 & 2/3 \end{pmatrix}}$$

The correlation is

$$r_{12} = \frac{s_{12}}{\sqrt{s_{11}s_{22}}} = \frac{-4/3}{\sqrt{(32/3)(2/3)}} = \frac{-4/3}{8/3} = \boxed{-\frac{1}{2}}$$

which is exactly $\cos \theta$ from above. So $R = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$. This shows that the length of a deviation vector is tied to the variance of that variable, and the angle between two deviation vectors is tied to their correlation.

Problem 3.6

$$X = \begin{pmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{pmatrix}$$

Column means:

$$\bar{x}_1 = \frac{-1 + 2 + 5}{3} = 2, \quad \bar{x}_2 = \frac{3 + 4 + 2}{3} = 3, \quad \bar{x}_3 = \frac{-2 + 2 + 3}{3} = 1$$

(a) Subtract the mean from each column to get $X - \mathbf{1}\bar{\mathbf{x}}'$:

$$X - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{pmatrix} = D$$

Is D full rank? Add up the rows:

$$(-3 + 0 + 3, 0 + 1 - 1, -3 + 1 + 2) = (0, 0, 0)$$

The rows add to the zero vector, so the rows are linearly dependent, and $\text{rank}(D) \leq 2$. Rows 1 and 2, $(-3, 0, -3)$ and $(0, 1, 1)$, are not multiples of each other, so $\text{rank}(D) = 2$. Since D is 3×3 , it is **not** full rank (full rank would be 3). This always happens for a mean-centered matrix because the columns of D each sum to zero.

(b) $S = \frac{1}{n-1} D' D = \frac{1}{2} D' D$.

$$D' D = \begin{pmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{pmatrix}$$

$$S = \frac{1}{2} \begin{pmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{pmatrix} = \boxed{\begin{pmatrix} 9 & -3/2 & 15/2 \\ -3/2 & 1 & -1/2 \\ 15/2 & -1/2 & 7 \end{pmatrix}}$$

Since D is not full rank, $D' D$ is singular, so

$$|S| = \boxed{0}.$$

Geometrically, $|S|$ measures the volume of the box formed by the deviation vectors. A value of 0 means the deviation vectors do not really span all 3 dimensions — they are squashed into a flat (2-dimensional) region, so there is no real 3-D spread in the data.

(c) Total sample variance = $\text{tr}(S) = 9 + 1 + 7 = \boxed{17}$.

Problem 3.9

$$X = \begin{pmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{pmatrix}$$

where $x_3 = x_1 + x_2$ (total score).

Column means:

$$\bar{x}_1 = 16, \quad \bar{x}_2 = 18, \quad \bar{x}_3 = 34$$

(and $34 = 16 + 18$, as expected.)

(a) Mean-corrected data matrix:

$$D = X - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{pmatrix}$$

In every row, column 3 is column 1 plus column 2 (e.g. row 1: $-4 + (-1) = -5$). So

$$\text{col}_1 + \text{col}_2 - \text{col}_3 = \mathbf{0}$$

which means

$$\mathbf{a}' = [a_1, a_2, a_3] = [1, 1, -1]$$

gives the linear dependence.

(b) $S = \frac{1}{n-1} D'D = \frac{1}{4} D'D.$

$$D'D = \begin{pmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{pmatrix}$$

$$S = \begin{pmatrix} 10 & 3 & 13 \\ 3 & 5/2 & 11/2 \\ 13 & 11/2 & 37/2 \end{pmatrix}$$

Since the columns of D are linearly dependent, $D'D$ (and S) is singular, so $|S| = \boxed{0}$.

Check $S\mathbf{a} = \mathbf{0}$ with $\mathbf{a} = (1, 1, -1)'$:

$$S\mathbf{a} = \begin{pmatrix} 10 + 3 - 13 \\ 3 + 2.5 - 5.5 \\ 13 + 5.5 - 18.5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

So $\mathbf{a}$ is an eigenvector of S with eigenvalue 0.

(c) Checking the original data: $12 + 17 = 29$, $18 + 20 = 38$, $14 + 16 = 30$, $20 + 18 = 38$, $16 + 19 = 35$. In each row $x_3 = x_1 + x_2$, confirming

$$1 \cdot x_1 + 1 \cdot x_2 + (-1) \cdot x_3 = 0,$$

i.e. $a_1 = 1$, $a_2 = 1$, $a_3 = -1$.

Problem 3.18

$$\bar{\mathbf{x}} = \begin{pmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{pmatrix}, \quad S = \begin{pmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.127 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{pmatrix}$$

(x_1 petroleum, x_2 natural gas, x_3 hydroelectric, x_4 nuclear.)

(a) Total consumption is $T = x_1 + x_2 + x_3 + x_4$, so $\mathbf{a} = (1, 1, 1, 1)'$.

Mean:

$$E[T] = 0.766 + 0.508 + 0.438 + 0.161 = \boxed{1.873}$$

Variance (= sum of all entries of S):

$$\text{Var}(T) = \mathbf{a}'S\mathbf{a}$$

$$\begin{aligned} &= (0.856+0.635+0.173+0.096)+(0.635+0.568+0.127+0.067)+(0.173+0.127+0.171+0.039)+(0.096+0.067+0.039+0.043) \\ &= 1.760 + 1.397 + 0.510 + 0.245 = \boxed{3.912} \end{aligned}$$

(b) Excess of petroleum over gas: $x_1 - x_2$, so $\mathbf{b} = (1, -1, 0, 0)'$.

Mean:

$$E[x_1 - x_2] = 0.766 - 0.508 = \boxed{0.258}$$

Variance:

$$\text{Var}(x_1 - x_2) = s_{11} + s_{22} - 2s_{12} = 0.856 + 0.568 - 2(0.635) = \boxed{0.154}$$

Covariance with the total T : $\mathbf{b}'S\mathbf{a}$. First find $\mathbf{b}'S$ (row 1 of S minus row 2 of S):

$$(0.856 - 0.635, 0.635 - 0.568, 0.173 - 0.127, 0.096 - 0.067) = (0.221, 0.067, 0.046, 0.029)$$

Then dot this with $\mathbf{a} = (1, 1, 1, 1)'$:

$$\text{Cov}(x_1 - x_2, T) = 0.221 + 0.067 + 0.046 + 0.029 = \boxed{0.363}$$

Problem 3.20

x_1	x_2	x_1	x_2	x_1	x_2
12.5	13.7	9.0	24.4	3.5	26.1
14.5	16.5	6.5	18.2	8.0	14.5
8.0	17.4	10.5	22.0	17.5	42.3
9.0	11.0	10.0	32.5	10.5	17.5
19.5	23.6	4.5	18.7	12.0	21.8
8.0	13.2	7.0	15.8	6.0	10.4
9.0	32.1	8.5	15.6	13.0	25.6
7.0	12.3	6.5	12.0		
7.0	11.8	8.0	12.8		

(x_1 = storm duration, x_2 = hours to clear snow, $n = 25$.)

(a) Using the summary statistics:

$$\bar{x}_1 = \frac{235.5}{25} = 9.42, \quad \bar{x}_2 = \frac{481.8}{25} = 19.272$$

$$\sum (x_{j1} - \bar{x}_1)^2 = 339.34 \Rightarrow s_{11} = \frac{339.34}{24} = 14.1392$$

$$\sum (x_{j2} - \bar{x}_2)^2 = 1493.7304 \Rightarrow s_{22} = \frac{1493.7304}{24} = 62.2388$$

$$\sum (x_{j1} - \bar{x}_1)(x_{j2} - \bar{x}_2) = 323.344 \Rightarrow s_{12} = \frac{323.344}{24} = 13.4727$$

Let $d = x_2 - x_1$. Then

$$\bar{d} = \bar{x}_2 - \bar{x}_1 = 19.272 - 9.42 = \boxed{9.852}$$

$$s_d^2 = s_{11} + s_{22} - 2s_{12} = 14.1392 + 62.2388 - 2(13.4727) = \boxed{49.4326}$$

(b) Now compute the differences directly, $d_j = x_{j2} - x_{j1}$:

1.2, 2.0, 9.4, 2.0, 4.1, 5.2, 23.1, 5.3, 4.8, 15.4, 11.7, 11.5, 22.5,

14.2, 8.8, 7.1, 5.5, 4.8, 22.6, 6.5, 24.8, 7.0, 9.8, 4.4, 12.6

Sum = 246.3, so

$$\bar{d} = \frac{246.3}{25} = \boxed{9.852}$$

which matches part (a) exactly. The sum of squared deviations is 1186.367, so

$$s_d^2 = \frac{1186.367}{24} = \boxed{49.4320}$$

Both methods give the same mean (9.852) and basically the same variance (49.4326 vs 49.4320, the small difference is just rounding). This makes sense because $\text{Var}(x_2 - x_1) = s_{11} + s_{22} - 2s_{12}$ should hold either way, whether we compute it from the summary statistics or straight from the differences.